NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) — path to success

RACE
Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2023
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED | **MATHEMATICS** | **M.M. 82**

REVISION ASSINGMENT # 06 (STRAIGHT LINE AND CIRCLE) TIME : 70 MIN.

SECTION–I

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. If a point P moves such that its distance from line $y = \sqrt{3}x - 7$ is same as its distance from $(3\sqrt{3}, 2)$, then the area of the curve, described by P, enclosed between the coordinate axes is (in square units) -

(A) $\frac{5\sqrt{3}}{2}$ (B) $5\sqrt{3}$ (C) $\frac{25\sqrt{3}}{2}$ (D) $25\sqrt{3}$

2. Let m_{AB} is slope of line joining the points A & B. Points $A_1(x_1, y_1)$, $A_2(x_2, y_2)$, ...& $A_n(x_n, y_n)$ are such that x_1, x_2, x_3 ..... x_n are in A.P. & y_1, y_2.....y_n are also in A.P., then $m_{A_1A_2} + m_{A_2A_3} + \ldots\ldots + m_{A_{(n-1)}A_n}$ is equal to -

(A) 0 (B) $(n-1)m_{A_1A_n}$ (C) $(n)\,m_{A_1A_n}$ (D) $(n^2)m_{A_1A_n}$

3. Let P(0, 3), Q(0, 5), R(x, 0), then the value of |x| for which $\angle PRQ$ is as large as possible :-

(A) $\sqrt{15}$ (B) $\sqrt{20}$ (C) $\sqrt{35}$ (D) $\sqrt{40}$

4. The mid point of chord AB of the circle $x^2 + y^2 - 6x - 4y + 3 = 0$ is the point (1, 1). If tangents are drawn at extremities A & B, then these tangents intersect at point -

(A) (1, 0) (B) (–1, 0) (C) (0, 1) (D) (1, –1)

5. The internal common tangents of the circles $x^2 + y^2 - 4x - 4y + 4 = 0$ and $x^2 + y^2 + 6x + 6y + 9 = 0$ are :

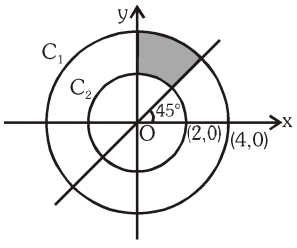
(A) $x - y = 2$ & $x + 2y = 3$ (B) $2x - 3y = 4$ & $x + 5 = 5$

(C) $x = 0$ & $y = 0$ (D) $x + 1 = 0$ & $y - 3 = 0$

6. If the radius of the circle touching $(x-2)^2 + y^2 = 1$ and the angle bisectors of the transverse common tangents of $(x \pm 2)^2 + y^2 = 1$ is $p \pm \sqrt{q}$, then $p + q$ is equal to -

(A) 8 (B) 9 (C) 10 (D) 11

7. As shown in figure C_1 & C_2 are two concentric circles with centres at origin and respective radii are 4, 2 units. If point (a, 3) lies inside the shaded region, then interval of a will be-



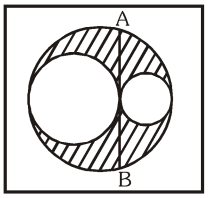
(A) $a > 0$ (B) $-\sqrt{7} < a < \sqrt{7}$

(C) $a < 3$ (D) $0 < a < \sqrt{7}$

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8. If the length of chord AB is 4 (as shown in adjacent figure), then find the area of shaded region -



(A) $\frac{\pi}{2}$ (B) π

(C) $\pi+1$ (D) 2π

9. If line $y = x - 2$ is normal to a circle, co-ordinates of its centre are such that abscissa is twice of ordinate & circle touches the x-axis then equation of the circle is -

(A) $x^2 + y^2 - 8x - 4y + 16 = 0$ (B) $x^2 + y^2 - 8x - 4y + 4 = 0$

(C) $x^2 + y^2 - 4x - 2y + 2y + 4 = 0$ (D) $x^2 + y^2 - 4x - 2y + 16 = 0$

10. Tangents drawn from (6, 4) to the circle $x^2 + y^2 - 4x - 4y + 4 = 0$ intersect the y-axis at A & B. AB is equal to

(A) 6 (B) 8 (C) 10 (D) 12

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

11. The diagonals of a square are along the pair of lines whose equation is $2x^2 - 3xy - 2y^2 = 0$. If (2, 1) is a vertex of the square, then the vertex of the square adjacent to it may be -

(A) (1, 4) (B) (–1, –4) (C) (–1, 2) (D) (1, –2)

12. ΔABC is an isosceles triangle in which A is (–1, 0), $\angle A = \frac{2\pi}{3}$, AB = AC and AB is along the x-axis. If BC = $4\sqrt{3}$, then for the line BC -

(A) the equation $x + \sqrt{3}y = 3$ (B) x-intercept = $\sqrt{3}$ (y-intercept)

(C) the equation $x - \sqrt{3}y = -3$ (D) area of triangle of $\Delta ABC = 4\sqrt{3}$

13. For the circles $S_1 = x^2 + y^2 - 4x - 6y - 12 = 0$ and $S_2 = x^2 + y^2 + 6x + 4y - 12 = 0$ and the line $L \equiv x + y = 0$

(A) L is common tangent of S_1 and S_2

(B) L is common chord of S_1 and S_2

(C) L is radical axis of S_1 and S_2

(D) L is Perpendicular to the line joining the centre of S_1 & S_2

14. C_1 & C_2 be 2 concentric circles with radii r_1 and r_2 ($r_1 < r_2$). If the tangents drawn from any point of C_2 to C_1 meet again C_2 at the ends of diameter of C_2, then -

(A) $r_2 = 2r_1$ (B) $r_2 = \sqrt{2}r_1$

(C) $r_2^2 < 2r_1^2$ (D) C_2 is director circle of C_1

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Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Question 15 to 16

Let ABC be a triangle whose vertex A is (3, 4). $L_1 = 0$ and $L_2 = 0$ are the internal angle bisectors of angles B and C respectively, where $L_1 \equiv x + y - 2 = 0$, $L_2 \equiv 2x + y - 5 = 0$. Let A_1 is the image of point A about line L_1.

On the basis of above information answer the following :

15. Co-ordinates of point A_1 are -

(A) (2, 1) (B) (–2, –1) (C) (2, –1) (D) (–1, –2)

16. Inradius of triangle ABC is -

(A) $\sqrt{10}$ (B) $2\sqrt{10}$ (C) $\frac{3}{2}\sqrt{10}$ (D) $\frac{5}{2}\sqrt{10}$

Paragraph for Question 17 to 18

Consider two circles $x^2 + y^2 = 1$ and $(x - 6)^2 + y^2 = 4$.

On the basis of above information, answer the following questions :

17. Length of direct common tangent of the circles is equal to-

(A) $\sqrt{24}$ (B) 5 (C) $\sqrt{35}$ (D) 6

18. Sum of the slopes of all the common tangents of the given circles is equal to-

(A) $3 + \sqrt{35}$ (B) 0 (C) $2\sqrt{3} + 2\sqrt{35}$ (D) $\sqrt{105}$

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SECTION–III

Numerical Grid Type (Single digit Ranging from 0 to 9) (4 Marks each, –1 for wrong answer)

1. A ray emitted from the point A(cosθ, sinθ) strikes a mirror at a variable point P, then reflects and passes through a point B(–sinθ, cosθ) such that incident and reflected rays are tangents to the circle $x^2 + y^2 = 1$, then locus of the variable point is $x^2 + y^2 = k$, then k is equal to

2. A ray of light incident at the point (–3,–2) gets reflected from the tangent at (0,–2) to the circle $x^2 + y^2 = 4$. The reflected ray touches the circle. The equation of the line along which the incident ray moved is $Jx - Ky + 46 = 0$.

The value of J + K is

3. The line $L_1 : 3x + 4y + 12 = 0$ is rotated through an angle of $\tan^{-1}\left(\frac{24}{7}\right)$ in anticlockwise about the point where it intersects the x-axis, line obtained in new position is denoted as $L_2 = 0$. If incentre of the triangle formed by $L_1 = 0$, $L_2 = 0$ and y-axis is $(4\lambda, 3k)$, then the value of $10k - 24\lambda$ is -

4. Let 'P' be any point inside the triangle OAB whose vertices are O(0, 0), A(3, 0) & B(3, 4). Symbol d(P, AB) represents distance of point P from the side AB. If P satisfies $d(P, OB) \le d(P, AB)$ and $d(P, OB) \le d(P, OA)$, then number of integral points inside the region in which P lies, is (where a point is said to be an integral point whose both co-ordinates are integers)

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SECTION–IV

Matrix Match Type (One or More than one option correct)

For each entry in **Column-I** , +2 If only the bubble(s) corresponding to all the correct matche(es) is (are) darkened, 0 In none of the bubbles is darkened, –1 In all other cases

1.

	Column-I	**Column-II**
(A)	If the circumference of the circle $x^2 + y^2 + 6x + 6y - b = 0$ is bisected by the circle $x^2 + y^2 - 4x + 2y + a = 0$, then $\|a + b\|$ is equal to	(P) 20
(B)	The point of intersection of the tangents of the circle $x^2 + y^2 = 10$, drawn at end points of the chord $x + y = 2$ is (a, b) then the value of $a^2 + b^2$ is equal to	(Q) 26
(C)	The intercept on the line $y = 2x$ by the circle $x^2 + y^2 - 5x = 0$ is AB. If the equation of the circle with AB as diameter is $x^2 + y^2 - ax - by = 0$ then the value of $2a + 9b$ is equal to	(R) 50
(D)	If the equation of the locus of the middle points of the chords of the circle $2x^2 + 2y^2 - 6x + 2y + 1 = 0$ that subtends an angle 120° at the centre is $x^2 + y^2 - ax - by + 2 = 0$ then $a^3 + b^3$ is equal to	(S) 42

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REVISION ASSINGMENT # 05 **PROBABILITY** **MATHEMATICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	B	C	D	C	A	B	B,C,D	A,B,C,D
	Q.	11	12	13	14	15					
	A.	A,B,C,D	C	C	A	C					
SECTION-II	Q.	1	2	3	4						
	A.	2.00	106.00	4.00	0.49						
SECTION-IV	Q.1	A	B	C	D						
		P	Q,T	P	S,T						

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